# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Master of Technology (M.Tech.)**
**SEM - I University Examination, Dec-Jan- 2013-14**
**M.Tech. (MECHANICAL -AMT)**
**ME 732: UNCONVENTIONAL MACHINING PROCESSES**

**Date: 04/01/2014** | **Time: 10:00 a.m. to 01:00 p.m.** | **Maximum Marks: 70**

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**Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheet.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw neat figures wherever if required.

## SECTION- I

**Q1 (a)** Discuss working of Plasma Arc Machining. **05**

**(b)** "AJM is not recommended to machine ductile material" Comment. **03**

**(c)** With the help of sketches, show the effect of standoff distance on (a) width of cut, (b) material removal rate in Abrasive Jet Machining. **03**

**Q2 (a)** With specific example, enlist the necessities that contribute the development of non-traditional machining processes and also make possible classification of non-traditional machining processes. **06**

**(b)** Explain the microscopic study of machined component by EDM process. **04**

**(c)** Why surface obtained in case of Chemical Machining of alloys is poor? Explain briefly. **02**

**OR**

**Q2 (a)** Briefly explain the concept of electrolysis. Explain with schematic diagram the working principal, characteristic and effects of electrolysis process on output parameters during Electrical Chemical Machining. **06**

**(b)** Explain the working principle of Magnetic Abrasive Finishing with the help of a neat sketch. Clearly show lines of magnetic force, magnetic equipotential lines, direction of pressure acting on the workpiece, direction of rotator motion and a semi-magnetic abrasive particle. **06**

**Q3 (a)** Define Electrical Discharge Machining? Briefly explain the principle of single discharge condition using neat sketch and effect of process parameters on surface finish. **08**

**(b)** With automatic electrode re-feed concept, explain the importance of servo-controlled system in Electro Discharge Machining. **04**

**OR**

**Q3 (a)** Sketch out the Waveforms of Relaxation power generator with its proper notations. Justify its charging and discharging conditions with circuit diagram used for Electrical Discharge Machining. **08**

**(b)** Brief out the importance of flushing system in Electro Discharge Machining. **04**

## SECTION – II

Q4 Write short note on

(a) Working of Ion Beam Machining process — 04

(b) Working of Ultrasonic Machining process — 04

(c) Effect of amplitude of vibration, frequency of vibration, grain size, and % of abrasive concentration, on MRR in ultrasonic machining process. — 03

Q5 (a) Prove that in constant voltage Electro Chemical Machining the highest accuracy may be achieved at highest productivity and the machining gap adjusts itself for any feed rate. — 08

(b) Brief out LASER generation process. — 04

**OR**

Q5 (a) Derive an expression for volumetric material removal rate for ultrasonic machining. — 08

(b) Explain the working of Electron Beam Machining Process with neat sketch. — 04

Q6 (a) Draw the schematic diagram and explain the working principle of Laser Machining with its important characteristics, advantages, limitations and fields of applications. — 08

(b) Elaborate Electrochemical Grinding material removal mechanism — 04

**OR**

Q6 (a) Derive expression of Material Removal Rate for ductile and brittle materials in Abrasive Jet Machining and discuss the condition for which AJM will produce equal MRR both for ductile and brittle material. — 08

(b) With the help of schematic diagram explain the roll of subsystems used in Water Jet Machining Process — 04

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